

No. Dosen : Pak Felix

Date 18/9/25

Statistika (Part. 1) SL (Introduction)

Exercises (hal 26)

1.1 List 20 situations involving uncertainty that happened with you yesterday

- ⇒
1. Alarm berbunyi tapi saya belum tentu langsung bangun.
 2. Listrik bisa saja mati saat pagi hari
 3. Air keran kadang lancar, kadang kecil alirannya.
 4. Sarapan tersedia atau habis lebih dulu
 5. Cuaca berubah-ubah antara cerah dan mendung
 6. Ojek online dtg cepat atau justru lama
 7. Jalan menuju pasar kadang lancar, kadang macet
 8. Harga sayuran diwarung tdk selalu sama
 9. Uang belanja bisa cukup atau kurang
 10. Sinyal Internet naik turun sepanjang hari
 11. Telepon dari keluarga masuk tiba-tiba
 12. Paket belanja online bisa dtg lebih cepat atau tertunda
 13. Pekerjaan rumah tangga selesai tepat waktu/tertunda
 14. Tamu dtg tanpa pemberitahuan sebelumnya.
 15. Hp / laptop bisa berfungsi normal atau tiba-tiba error
 16. Masakan kdg cukup, kdg kurang utk semua anggota keluarga.
 17. Suasana rumah bisa tenang, bisa juga ramai
 18. Kabar dari teman kadang baik, kdg mengejutkan
 19. Badan terasa segar dipagi hari, tapi bisa lelah di sore hari
 20. Tidur malam bisa nyenyak, bisa juga gelisah.

1.2 Name 10 random variables that you observed or dealt with yesterday.

1. Jumlah menit menunggu angkot/bus
2. Waktu tempuh perjalanan ke kampus (dalam menit)
3. Jumlah soal yg diberikan dosen utk latihan.
4. Banyaknya uang yg saya keluarkan di kantin.
5. Banyaknya pesan masuk di grup WhatsApp kelas.
6. Jumlah halaman materi yg sempat saya baca
7. Durasi listrik menyala di rumah (jam)
8. Jumlah teman yg hadir di kelas
9. Banyaknya tgs yg harus dikumpulkan
10. Jam tidur saya (pukul berapa akhirnya tidur).

- 1.4 (a) koin
- kepala $\rightarrow$ main game ($P = 1/2$)
 - ekor $\rightarrow$ nonton video ($P = 1/2$)
 - berdiri $\rightarrow$ tugas ($P = 0/1$)
 - melayang $\rightarrow$ belajar ($P = 0$)

1.3 Name 5 Stochastic processes that played a role in your actions yesterday.

1. Cuaca sepanjang hari - kondisi berubah secara acak dari cerah, mendung.
2. Pesan atau notifikasi di Hp - muncul secara acak dim turun waktu tertentu.
3. Waktu kedatangan paket kurir - tidak pasti kpn tiba di rumah.
4. Laju konsumsi baterai hp - bergantung pd pemakaian yg bervariasi sepanjang hari.
5. Gangguan listrik / mati lampu - bisa muncul secara acak.

1.4 (a). Probabilitas 0 $\rightarrow$ koin berdiri dtisi, koin menggantung di udara.

- Probabilitas 1 $\rightarrow$ hasil "heads atau tails".
- Probabilitas antara 0 dan 1 $\rightarrow$ hasil "heads" saja atau "tails" saja.

(b) "Watch a video" terjadi jika tails keluar

- Untuk sebuah fair coin - kita definisikan bahwa peluang heads = peluang tails = $1/2$
- jadi probabilitas "Watch a video" = 0.5

ini langsung membantu mendefinisikan koin adil (fair coin)

Fair coin = koin dimana peluang heads dan tails sama besar, yaitu $1/2$ utk masing².

1.5 Perangkat Lunak

- Apakah bisa diprediksi? Bisa, dgn data historis dan model statistik / time series
- Data yg perlu dikumpulkan: jumlah defect ditemukan & diperbaiki per hari, effort testing, backlog defect, perubahan kode, coverage pengujian.
- Predictor: hari ke-n, effort testing, backlog defect, jumlah commit / perubahan kode.
- Response: jumlah defect yg ditemukan per hari.

1.6 Toko Komputer

- Data yg perlu dikumpulkan: jumlah produk (SKU) yg distok, volume penjualan per produk, harga jual, biaya brg (COGS), biaya operasional
- Predictor: jumlah SKU, stok per produk, harga, biaya, volume penjualan, faktor promosi / musiman
- Response: keuntungan bulanan (monthly profit).

No. Kamis

Date 25/9/25

Dosen: Pak Felix

Probability (Pert. 2 SL)

- (2.1) Out of six computer chips, two are defective. If two chips are randomly chosen for testing (without replacement), compute the probability that both of them are defective. List all the outcomes in the sample space.

↳ penyelesaian:

Jumlah chip = 6

rusak^(D) = 2, bagus (G) = 4

D = Defective (rusak)

G = Good (bagus)

chip rusak : D1, D2

chip bagus : G1, G2, G3, G4

Sample space : memilih 2 chip dari 6 → jumlah kombinasi

$$\binom{6}{2} = \frac{6 \times 5}{2} = \frac{30}{2} = 15$$

D1D2,

→ pasangan yg berisi kedua chip

D1G1, D1G2, D1G3, D1G4, maka, peluang = (jumlah hasil memenuhi)

D2G1, D2G2, D2G3, D2G4, (jumlah total hasil)

G1G2, G1G3, G1G4,

$$= \frac{1}{15} \approx 0,0667 = 6,67\%$$

G2G3, G2G4,

G3G4,

- (2.2) Suppose that after 10 years of service, 40% of computers have problems with motherboards (MB), 30% have problems with hard drives (HD), and 15% have problems with both MB and HD. What is the probability that a 10-year old computer still has fully functioning MB and HD?

↳ penyelesaian:

$$P(\text{MB}) = 0.40 \quad (40\%)$$

$$P(\text{HD}) = 0.30 \quad (30\%)$$

$$P(\text{MB} \cap \text{HD}) = 0.15 \quad (15\%)$$

↳ rumus union:

$$P(\text{MB} \cup \text{HD}) = P(\text{MB}) + P(\text{HD}) - P(\text{MB} \cap \text{HD})$$

$$= 0.40 + 0.30 - 0.15$$

$$= 0.55 \quad (55\% \text{ alami masalah pd MB dan HD})$$

maka, probabilitas komputer masih berfungsi baik:

$$(\text{tidak MB, tidak HD}) = P(\text{baik}) = 1 - 0.55 = 0.45$$

- 2.3 A new computer virus can enter the system through e-mail or through the Internet. There is a 30% chance of receiving this virus through e-mail. There is a 40% chance of receiving it through the Internet. Also, the virus enters the system simultaneously through e-mail and the Internet with probability 0.15. What is the probability that the virus does not enter the system at all?

1) Penyelesaian :

$$\rightarrow P(\text{virus lewat email}) = 30\% \rightarrow 0,30$$

$$P(\text{virus lewat internet}) = 40\% \rightarrow 0,40$$

$$P(\text{keduanya}) = 0,15$$

Probabilitas virus masuk ($E \cup I$)

$$\begin{aligned} P(E \cup I) &= P(E) + P(I) - P(E \cap I) \\ &= 0,30 + 0,40 - 0,15 \\ &= 0,55 \end{aligned}$$

Jadi, probabilitas tidak masuk sama sekali :

$$1 - 0,55 = 0,45 \text{ (45\%)}$$

- 2.4 Among employees of a certain firm, 70% know Java, 60% know Python, and 50% know both languages. What portion of programmers.

(a) does not know Python?

(b) does not know Python and does not know Java?

(c) knows Java but not Python?

(d) knows Python but not Java?

(e) If someone knows Python, what is the probability that he/she know Java too?

(f) If someone knows Java, what is the probability that he/she knows Python too?

⇒ Dik : $P(J) = 70\% \rightarrow 0,70$ (tahu Java)

$$P(Py) = 60\% \rightarrow 0,60 \text{ (tahu Python)}$$

$$P(J \cap Py) = 50\% \rightarrow 0,50 \text{ (tahu keduanya).}$$

pakai cth basis 100 karyawan.

$$\begin{aligned} \Rightarrow \text{tahu Java saja} &= P(J) - P(J \cap Py) & \text{tahu Python saja} &= P(Py) - P(J \cap Py) \\ &= 0,70 - 0,50 & &= 0,60 - 0,50 \\ &= 0,20 \text{ (20\%)} & &= 0,10 \text{ (10\%)} \end{aligned}$$

$$\Rightarrow \text{tahu keduanya} = 0,50 \rightarrow 50\%$$

$$\Rightarrow \text{tdk tau Java \& Python}$$

$$= 1 - (0,20 + 0,10 + 0,50)$$

$$= 1 - 0,80$$

$$= 0,20 \rightarrow 20\%$$

$$\begin{aligned} a. &= 1 - P(Py) \\ &= 1 - 0.60 \\ &= 0.40 \text{ (40 org)} \end{aligned}$$

$$\begin{aligned} b. &= 1 - P(J \cup Py) \rightarrow 0.70 + 0.60 - 0.50 \\ &= 1 - 0.80 \\ &= 0.20 \text{ (20 org)} \end{aligned}$$

$$c. 0.20 \text{ (20 org).}$$

$$d. 0.10 \text{ (10 org)}$$

$$e. P(\text{Java} | \text{Python}) = \frac{P(\text{both})}{P(\text{Python})} = \frac{0.50}{0.60} = \frac{5}{6} \approx 0.8333 = 83.33\%$$

$$f. P(\text{Python} | \text{Java}) = \frac{P(\text{Both})}{P(\text{Java})} = \frac{0.50}{0.70} = \frac{5}{7} \approx 0.7143 = 71.43\%$$

2.5 A computer program is tested by 3 independent tests. When there is an error, these tests will discover it with probabilities 0.2, 0.3, and 0.5, respectively. Suppose that the program contains an error. What is the probability that it will be found by at least one test?

Penyelesaian:

- Probabilitas ditemukan oleh tes 1 = 0.2
- tes 2 = 0.3
- tes 3 = 0.5

↳ Probabilitas tdk ada tes yg menemukan error

$$= (1 - 0.2) (1 - 0.3) (1 - 0.5)$$

$$= 0.8 \times 0.7 \times 0.5$$

$$= 0.28$$

maka probabilitas minimal satu tes menemukan error:

$$= 1 - 0.28 = 0.72 \text{ (72\%)}$$

2.6 Under good weather conditions, 80% of flights arrive on time. During bad weather, only 30% of flights arrive on time. Tomorrow, the chance of good weather is 60%. What is the probability that your flight will arrive on time?

Penyelesaian:

$$\text{Dik: } P(\text{OnTime} | \text{Good}) = 80\% \rightarrow 0.80$$

$$P(\text{Good}) = 60\% \rightarrow 0.60$$

$$P(\text{OnTime} | \text{Bad}) = 30\% \rightarrow 0.30$$

$$P(\text{Bad}) = 1 - 0.60 = 0.40$$

$$P(\text{OnTime}) = P(\text{OnTime} | \text{Good}) \cdot P(\text{Good}) + P(\text{OnTime} | \text{Bad}) \cdot P(\text{Bad})$$

$$= (0.80 \times 0.60) + (0.30 \times 0.40)$$

$$= 0.48 + 0.12$$

$$= 0.60 \text{ (60\%)}$$

2.7 A system may become infected by some spyware through the internet or e-mail. Seventy percent of the time the spyware arrives via the internet, thirty percent of the time via e-mail. If it enters via the internet, the system detects it immediately with probability 0.6. If via e-mail, it is detected with probability 0.8. What percentage of times is this spyware detected?

$$\Rightarrow P(\text{detected}) = P(\text{Detected} | \text{Internet}) \cdot P(\text{Internet}) + P(\text{Detected} | \text{Email}) \cdot P(\text{Email})$$

$$\begin{aligned} (\text{Internet}) &\rightarrow 0.6 \times 0.7 = 0.42 & \text{* Dik: } 70\% \text{ Spyware Internet} &\rightarrow P(0.7) \\ (\text{Email}) &\rightarrow 0.8 \times 0.3 = 0.24 & 30\% \text{ lewat email} &\rightarrow P(0.3) \\ \text{Total} &\cdot P(\text{Detected}) = 0.42 + 0.24 = 0.66 \uparrow 66\% \end{aligned}$$

2.8 A shuttle launch depends on three key devices that may fail independently of each other with probabilities 0.01, 0.02, and 0.02, respectively. If any of the key devices fails, the launch will be postponed. Compute the probability for the shuttle to be launched on time, according to its schedule.

$$\Rightarrow \text{Dik: Perangkat 1} \rightarrow 0.01 \rightarrow 1 - 0.01 = 0.99$$

$$2 \rightarrow 0.02 \rightarrow 1 - 0.02 = 0.98$$

$$3 \rightarrow 0.02 \rightarrow 1 - 0.02 = 0.98$$

Dit: Probabilitas peluncuran sesuai jadwal (tdk ada gagal)

$$\Rightarrow P(\text{On Time}) = 0.99 \times 0.98 \times 0.98$$

$$= 0.950796 \approx 0.951 \approx 95.1\%$$

2.9 Successful implementation of a new system is based on three independent modules. Module 1 works properly with probability 0.96. For modules 2 dan 3, these probabilities equal 0.95 and 0.90. Compute the probability that at least one of those three modules fails to work properly.

$$\Rightarrow \text{Dik: } P(M_1 \text{ bekerja}) = 0.96$$

$$P(M_2 \text{ bekerja}) = 0.95$$

$$P(M_3 \text{ bekerja}) = 0.90$$

Dit: Probabilitas mis satu gagal

$$\text{Jwb: } P(\text{semua berfungsi}) = 0.96 \times 0.95 \times 0.90$$

$$= 0.8208$$

$$P(\geq 1 \text{ gagal}) = 1 - 0.8208$$

$$= 0.1792 \approx 17.92\%$$

Soal dari materi modulSoal 1

Ada kemungkinan 30% virus masuk ke sistem melalui email dan 40% melalui internet.

Kelain itu, virus dpt memasuki sistem secara bersamaan melalui email dan internet dgn probabilitas 15%. Berapa kemungkinan virus tsb tdk masuk ke sistem sama sekali?

$$\Rightarrow \text{Dik: } P(E) = 0.30$$

$$P(I) = 0.40$$

$$P(E \cap I) = 0.15$$

Dit: P virus tdk masuk sama sekali?

$$\text{Jwb: } P(E \cup I) = 0.30 + 0.40 + 0.15$$

$$= 0.55$$

$$P(\text{tdk masuk}) = 1 - 0.55$$

$$= 0.45 \text{ | } 45\%$$

Soal 2

Setelah digunakan selama 10 tahun, 45% komputer mengalami masalah pd motherboard (MB), 30% mengalami masalah pd harddisk (HD), dan 17% mengalami masalah pd keduanya, yaitu MB dan HD. Bp peluang komputer yg berumur 10 tahun masih berfungsi penuh tanpa mengalami masalah pd MB dan HD?

$$\Rightarrow \text{Dik: } P(MB) = 0.45$$

$$P(HD) = 0.30$$

$$P(MB \cap HD) = 0.17$$

Dit: P tdk bermasalah sama sekali?

$$\text{Jwb: } P(MB \cup HD) = 0.45 + 0.30 - 0.17$$

$$= 0.58$$

$$P(\text{tdk sama sekali}) = 1 - 0.58$$

$$= 0.42 \text{ (42\%)}$$

Date 9/10/25

Statistika Pert. 4 SL (Discrete Random Variables and Their Distributions)

3.1 A computer virus is trying to corrupt two files. The first file will be corrupted with probability 0.4. Independently of it, the second file will be corrupted with probability 0.3.

(a) Compute the probability mass function (pmf) of X , the number of corrupted files.

$$\Rightarrow P(A_1) = 0.4$$

$$P(A_2) = 0.3$$

X = bisa bernilai 0, 1 atau 2

$$P(X = x)$$

$\rightarrow P(X = 0)$ tdk ada file rusak

$$= (1 - 0.4)(1 - 0.3)$$

$$= 0.6 \times 0.7 = 0.42$$

$\rightarrow P(X = 1)$ satu file rusak

$$= \text{file 1 rusak, 2 tdk} : 0.4 \times 0.7 = 0.28$$

$$= \text{file 1 tdk, 2 rusak} : 0.6 \times 0.3 = 0.18$$

$$\underline{0.46} +$$

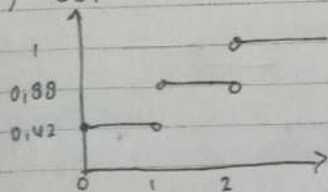
$\rightarrow P(X = 2)$ kedua rusak

$$= 0.4 \times 0.3 = 0.12$$

Fungsi pmf:

X	0	1	2
$P(X=x)$	0.42	0.46	0.12

(b) CDF



- 3.14 An internet service provider charges its customers for the time of the internet use rounding it up to the nearest hour. The joint distribution of the used time (X , hours) and the charge per hour (Y , cents) is given in the table below.

→

$P(x,y)$		x			
		1	2	3	4
y	1	0	0.06	0.06	0.10
	2	0.10	0.10	0.04	0.04
	3	0.40	0.10	0	0

Each customer is charged $Z = X \cdot Y$ cents, which is the number of hours multiplied by the price of each hour. Find the distribution of Z .

⇒ $Z = P(Z = z)$

1 $P(Z = 1) = P(X = 1, Y = 1) \rightarrow 0$

2 $P(2, 1) + P(1, 2) \rightarrow 0.06 + 0.10 = 0.16$

3 $P(3, 1) + P(1, 3) \rightarrow 0.06 + 0.40 = 0.46$

4 $P(4, 1) + P(2, 2) \rightarrow 0.10 + 0.10 = 0.20$

6 $P(3, 2) + P(2, 3) \rightarrow 0.04 + 0.10 = 0.14$

8 $\rightarrow 0.04$

9 $\rightarrow 0$

12 $\rightarrow 0$

1.00

Nilai paling mungkin biaya total Z adalah 3 sen
kemungkinan tertinggi 0.46.

- 3.15 Let X dan Y be the number of hardware failures in two computer labs in a given month. The joint distribution of X dan Y is given in the table below.

$P(x,y)$		x		
		0	1	2
y	1	0.52	0.20	0.04
	2	0.14	0.02	0.01
	3	0.06	0.01	0

- (a) compute the probability of at least one hardware failure.

⇒ Pstidaknya satu kerusakan = $1 - P(X = 0, Y = 0)$

$= 1 - 0.52$

$= 0.48$

- (b) from the given distribution, are X dan Y independent? Why or why not?

⇒ for $X \rightarrow 0.52 + 0.14 + 0.06 = 0.72$ for $Y \rightarrow 0.52 + 0.20 + 0.04 = 0.76$

$0.20 + 0.02 + 0.01 = 0.23$

$0.14 + 0.02 + 0.01 = 0.17$

$0.04 + 0.01 + 0 = 0.05$

$0.06 + 0.01 + 0 = 0.07$

$P(X = 0, Y = 0) = 0.52$

$P_X(0) \cdot P_Y(0) = 0.72 \times 0.76 = 0.5472 \rightarrow \neq$ tdk independen.

Statistika SL (Pert. 6) Continuous Distributions

4.1 Umur (lifetime) dim tahun, dari suatu komponen elektronik merupakan peubah acak kontinu dengan fungsi kepadatan peluang:

$$f(x) = \begin{cases} \frac{k}{x^4}, & \text{utk } x \geq 1 \\ 0, & \text{utk } x < 1 \end{cases}$$

tentukan nilai k , fungsi distribusi (CDF), $F(x)$ dan peluang bahwa umur komponen lebih dari 2 tahun yaitu $P(X > 2)$

Dik: Dit: Nilai k

Fungsi distribusi kumulatif

Peluang

a. menentukan nilai k

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$f(x) = 0$ utk $x < 1$, maka:

$$\int_1^{\infty} \frac{k}{x^4} dx = 1$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$n = -4 \Rightarrow n+1 = -3$$

Hitung integral:

$$\int_1^{\infty} x^{-4} dx = \left[\frac{x^{-3}}{-3} \right]_1^{\infty} = 0 - \left(-\frac{1}{3} \right) = \frac{1}{3}$$

$$k \left(\frac{1}{3} \right) = 1 \Rightarrow \boxed{k = 3}$$

fungsi distribusi kumulatif didefinisikan sebagai:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

$$b. f(x) = \begin{cases} 0, & x < 1 \\ \int_1^x \frac{3}{t^4} dt, & x \geq 1 \end{cases}$$

$$f(x) = \begin{cases} 0 \end{cases}$$

$$\text{Hitung: } \int_1^x 3t^{-4} dt = 3 \left[\frac{t^{-3}}{-3} \right]_1^x = -(t^{-3})_1^x = 1 - \frac{1}{x^3}$$

$$\text{Jadi, } \boxed{F(x) = 1 - \frac{1}{x^3}, x \geq 1}$$

$$c. P(X > 2) = 1 - P(X \leq 2) = 1 - F(2)$$

$$b. F(2) = 1 - \frac{1}{2^3} = 1 - \frac{1}{8} = \frac{7}{8}$$

$$\Rightarrow P(X > 2) = 1 - \frac{7}{8} = \frac{1}{8} = 0,125$$

$$44 \quad f(x) = \begin{cases} K - \frac{x}{50} & ; 0 < x < 10 \\ 0 & ; \text{otherwise} \end{cases}$$

$$* \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^0 f(x) dx + \int_0^{10} f(x) dx + \int_{10}^{\infty} f(x) dx = 1$$

$$\int_0^{10} (K - \frac{x}{50}) dx = 1$$

$$\int_0^{10} K dx - \int_0^{10} \frac{x}{50} dx = 1$$

$$Kx \Big|_0^{10} - \left[\frac{1}{100} x^2 \right]_0^{10} = 1$$

$$[K \cdot 10 - K \cdot 0] - \left[\left(\frac{1}{100} \cdot 100 \right) - \left(\frac{1}{100} \cdot 0 \right) \right] = 1$$

$$10K - 1 = 1$$

$$K = \frac{1}{5}$$

$$* P(X < 5) = \int_0^5 (K - \frac{x}{50}) dx$$

$$= \left[Kx - \frac{x^2}{100} \right]_0^5$$

$$= \left[\frac{1}{5} \cdot 5 - \frac{25}{100} \right] - \left[\frac{1}{5} \cdot 0 - \frac{0^2}{100} \right]$$

$$= \frac{3}{4} = 0,75$$

$$* E(X) = \int x f(x) dx$$

$$= \int_0^{10} x \left(\frac{1}{5} - \frac{x}{50} \right) dx$$

$$= \left[\frac{x^2}{10} - \frac{x^3}{150} \right]_0^{10}$$

$$= 10 - \frac{100}{15} - (0)$$

$$= \frac{50}{15} = \frac{10}{3} = 3,333$$